

University of Gondar
College of Informatics
Department of Computer Science
Image processing and Computer Vision Group Project

Deadline: April 28, 2018 E.c

Use getnet6202@gmail.com to submit

General Instructions

- Form groups consisting of no more than ten (10) students per group.
- Each group must select a unique project title in the field of Image Processing and Computer Vision. The project may focus on image detection, classification, or a combination of both.

Project Development Environment

- The project should be implemented using suitable environments such as:
 - Google Colab, or
 - A locally installed Anaconda environment.

Dataset Preparation

- Use an image dataset containing at least two or more classes.
- The dataset may be obtained from multiple sources, including:
 - Public repositories
 - Manually captured images (e.g., using mobile phones)
 - Locally available datasets
- Perform appropriate data preprocessing, including:
 - Data augmentation
 - Data splitting (training, validation, and testing sets)
 - Class balancing and other necessary preprocessing techniques
- Create and present a bar chart illustrating the dataset size for each class:
 - Before data splitting

- After data splitting

Algorithm Selection

- Each group must select one unique algorithm for model training, validation, and testing.
- The algorithm may be chosen from, but is not limited to:
 - VGGNet and its variants
 - ResNet and its variants
 - Vision Transformers and their variants
 - MobileNet and its variants

Model Training and Evaluation

- Load the training, validation, and testing datasets.
- Train the model for up to 150 epochs, incorporating early stopping to prevent overfitting.
- Save the trained model in .h5 format.
- Plot and analyze:
 - Training and validation loss curves
 - Training and validation accuracy curves
- Evaluate the model using the testing dataset and present:
 - Classification report
 - Confusion matrix
 - ROC curve
- *(Optional but recommended):* Apply advanced techniques such as:
 - Model pruning
 - Overfitting reduction strategies
 - Any additional methods to improve model performance

Report Submission Guidelines (no more than 20 pages including the references)

Cover Page

The cover page must include:

- University name and details
- Group member details
- Project title
- Submission date

Main Body

- Abstract (max 250 words and include no more than five keywords)
- Introduction (overview of the project max one and half page)
- Methodology (max 6 pages):
 - Dataset preparation (sources and preprocessing steps)
 - Proposed model architecture(s)
 - Model evaluation metrics (accuracy, precision, recall, and ROC curve)
- Experiments(max 6 pages):
 - Development environment
 - Model training strategy
 - Results and discussion
 - Including plots of confusion matrix, classification report, loss and accuracy curve and roc curve
- Conclusion and Recommendations
- References